Major-Comment Response to Two Referee Reports

AI Exposure or Occupational Composition?
Constructed Measures and Early-Career Employment

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Scope of this revision. This document responds to every numbered major comment in the two reports supplied on September 5, 2026. In accordance with the staged revision requested by the author, secondary copyediting and presentation comments are catalogued for the next pass rather than represented as complete. All post-outcome analyses are labeled exploratory; the frozen v1.1 design and results remain intact.

Revision in one paragraph

We accepted the central diagnosis and rewrote the paper around the result the CPS can establish. The corrected Q5–Q1 beta coefficient is −.1346, but it falls to −.0315 after broad-family-specific post changes are absorbed and to −.0317 after family-specific monthly paths are absorbed. The paired movement is detectable, although the conditional estimand changes and retains only 30 percent of baseline target information. Clean service exclusions do not support a food-service mechanism. We now report a primary MDE_{80} of .1244 and paired MDEs for every architecture comparison; all seven paired Q5–Q1 intervals include zero. The title, abstract, introduction, results, and conclusion no longer claim general architecture heterogeneity or AI-specific displacement. The revised contribution is a composition-centered empirical finding joined to a framework that separates implementation error, measurement, construct, support, estimand, representation, and reparameterization changes.

Response to Referee 1

M1. Precision and comparative language

Response. Accepted. The main text now reports the realized primary MDE_{80} of .1244 log points next to the −.1311 frozen estimate. A consolidated paired table reports MDEs from .0609 to .1689 for beta against the three AIOE implementations, alpha, broad, Webb AI, and OECD. Every paired Q5–Q1 interval contains zero. Throughout, we now write "the design does not detect a difference" rather than "the conclusion does not generalize," and never infer equivalence.

We also diagnosed the prospective-to-realized SE gap. A sharp historical simulation preserves each donor month's full contemporaneous cross-occupation residual vector with a global sign. Its mean SE remains .01217, so it closes none of the gap to .04441. We report this as a rejection of one proposed mechanism, not a decomposition or validation of the planning simulation. See Sections 1, 4, and 6; Main Table 3 and Figure 4; Appendix D and K.

M2. Effective clustering and CPS rotation

Response. Addressed within the available survey design. Young-cell respondent equivalents have p10/median/p90 of 0/2/16; 26.3 percent are zero and 69.9 percent below five. Quarterly aggregation gives $-.13103$ on the frozen and $-.13454$ on the repaired calendar, nearly identical to monthly estimates and SEs. A cross-occupation time-score covariance sensitivity increases the SE to .04909 at a 12-month Newey–West lag, with normal interval $[-.2273, -.0349]$.

The main text now explains that occupation clustering already permits arbitrary serial dependence within occupation but is not a complete CPS design-based variance. The additional HAC targets common monthly and nearby-month score covariance across occupations. The extract begins in 2017, so literal 2015–19 pseudo-breaks are unavailable. We run all feasible 2017–19 dates and emphasize 12 balanced dates; none approaches $-.1311$. Their 1/13 empirical tail is explicitly a resolution floor in a dependent, post-outcome diagnostic, not a randomization p-value. See Section 4 and Appendix D.

M3. Regression analogue of the permutation, monotonicity, and stable tails

Response. This result now determines the paper. On the corrected calendar, SOC2-by-young-by-post terms reduce $-.1346$ to $-.0315$; SOC2-by-young-by-month paths give $-.0317$. Paired baseline-to-conditioned movements are approximately .103 and their intervals exclude zero. We do not call this a causal decomposition: conditioning changes the estimand, only four of 22 families span Q1 and Q5, and target information falls to 30 percent.

We add a three-restriction test of a common Q2–Q5 coefficient ($p = .119$) and a least-favorable max-t monotonicity assessment ($p = .393$). Neither equality nor monotonicity is established. The 46 always-Q1 and 18 always-Q5 occupations yield $-.2120$ with interval $[-.4131, -.0109]$, but only 15.0 effective information occupations and a 46.7-percent top-five share. See Section 5, Main Table 2 and Figure 2, and Appendix C/E.

M4. Influence concentration and service recovery

Response. Influence is now a headline diagnostic rather than summarized by the absence of sign reversal. Deleting the five occupations with largest absolute prior LOCO movements gives $-.1011$. Deleting ten or twenty gives $-.1522$ and $-.1555$, showing offsetting signed influence. A symmetric signed-tail trim gives $-.1281$ and continuous Huber down-weighting $-.1314$. These data-adaptive rows are labeled diagnostics, not preferred estimators.

We replaced the coarse industry proxy with occupation-major-group exclusions. Dropping Q1 food-service occupations gives $-.1196$, an 8.8-percent attenuation. Dropping broader in-person services gives $-.1370$ or $-.1387$. We therefore reject the proposed strong food-service mechanism while accepting that the Q1 trajectory and individual occupations matter. See Section 5 and Appendix E/J.

M5. CPS population controls

Response. The respondent-equivalent result is promoted and defined operationally as cells reconstructed from observed records without final CPS weights. It is not a common rescaling of weighted stocks. On the repaired calendar it is $-.1348$, versus $-.1346$ under official weights. Ending in December 2024 gives $-.1056$ and $-.1135$, respectively.

BLS does not publish age-by-occupation bridge weights for the January 2025 revision; its experimental factors are aggregate. We therefore do not fabricate a counterfactual micro-weight series. The raw official-weight contrast becomes slightly less negative between December 2024 and January 2025, and the official 2025–26 era is not distinguishable from 2023–24. The documentation, bounded inference, and refusal are now explicit. See Section 7 and Appendix F.

M6. Selected architecture space

Response. The revision includes a repository-bounded candidate census with source-integrity, construct, mapping/support, and representation status. It records six selected implementations from two core families, two operationally admitted external implementations from two other families, one scope failure, and multiple pending or locally unverified candidates. It does not claim a global count.

The weighted eigenvalue spectrum confirms the referee's concern: PC1 explains 80.04 percent and the first two components 96.11 percent of common-support score variance. The IUT is demoted to a predeclared marginal sign statement over a dependent set; the failed simultaneous band remains visible. Firm- and demand-side measures are not forced into occupation CPS without a defensible unit bridge. See Section 2 and Appendix A.

M7. Inconsistent confidence intervals

Response. Corrected. The only primary frozen interval is the 9,999-draw $[-.217075, -.045073]$. We trace the earlier variants to different finite 999-multiplier sets; the minimum-count row additionally changed support. Pair-specific member intervals are labeled and are not presented as primary. A consistency audit checks the canonical coefficient and interval. See Section 4 and Appendix D.

M8. Relationship to Brynjolfsson, Chandar, and Chen

Response. We implement the reproducible public component of BCC: GPT-4 beta with top two employment-weighted quintiles against the bottom three. In CPS this gives $-.0728$ with interval $[-.1240, -.0216]$, compared with the corrected YAX Q5–Q1 coefficient $-.1346$. Applying that grouping across eight constructions yields native-support estimates from $-.0817$ to $-.0096$. On literal common support, beta differs detectably from OECD under this grouping but not from the other six alternatives.

We label this a bridge, not a replication. The ADP outcome, employer panel, job-title mapping, firm controls, hiring, and separation margins are proprietary or outside CPS. See Sections 2 and 6 and Appendix E.

M9. January 2023 and adoption

Response. We supplement the binary coefficient with the existing event-time profile, balanced pre-AI pseudo-breaks, and joint post-era coefficients. We do not introduce a new adoption series in this revision. A national adoption series interacted with fixed exposure would largely reparameterize exposure-by-time dynamics already shown. BTOS would require an occupation–industry ecological bridge and still would not identify exogenous adoption.

This is a principled scope decision, not a claim that adoption is irrelevant. The binary term is now called a chronology benchmark, not a causal shock. A separate adoption design remains future work. See Section 7 and Appendix F.

M10. F/G reparameterization

Response. Accepted. F/G is removed from the main results and retained in Appendix I as a coordinate transformation. The main text leads with the representative AIOE-plus-beta and direct D, S models. The task-family centroid's effective raw-unit weights are reported as $4.9057D + 1.8104S$, so the embedded S-to-D weight is .3690. We no longer treat a confidence interval crossing zero after component deletion as evidence of a detected coefficient change.

M11. Mobility, entry, and uncertainty

Response. The employed-to-employed switch analysis is removed from the paper's main contribution. The appendix now separates the entrant-destination allocation among people observed entering employment from any employment-finding probability. Its beta Q5–Q1 coefficient is $-.0888$ with interval $[-.2787, .1011]$ and $p = .383$. Immediate reversals are not treated as a validated coding-error rate, so no purported attenuation correction is applied; the appendix also reconciles the prior 9.865-percent statistic with the present sample's 10.336-percent raw conditional rate.

The revised mobility audit reports a 20.13-percent family-balanced pairwise disagreement rate and proves that beta adds no new zero-threshold conflict when its task-family endpoints agree. On identical 98.305-percent represented support, realized conflict is 53.2819 percent and the benchmark is 52.3227 percent, a 0.9592-point gap. The declared household-cluster SE is 0.1979 point for the realized rate and 0.0776 point for the gap, versus a 0.00172-point Monte Carlo SE for the benchmark mean. The all-support accounting bound for the gap is [0.1521,1.8469] points. A second valid repair rule changes the benchmark by only −0.00157 point, but neither rule is a canonical uniform draw over feasible derangements. The main text draws no hiring conclusion from switches. See Appendix G/H.

M12. Generated-covariate validation

Response. Appendix A now specifies a validation design: occupation–task units stratified by family and architecture disagreement; separate labels for capability, realized time savings with and without software, adoption, and task importance; blinded independent experts plus audited use logs where feasible; repeated labels; held-out occupations; and versioned task/model/mapping records. We provide a Fisher-z power formula and an illustrative independent-unit calculation while emphasizing that clustering, rater noise, sampling weights, and target correction determine the actual sample size. We do not invent one universal number.

Response to Referee 2

M1. Framework separating measurement from the question

Response. Accepted and used to reorganize the paper. Main Table 1 distinguishes implementation repair, same-construct measurement, construct change, population/support change, estimand change, representation change, and pure reparameterization. The paper states positive-affine rank/bin invariance, piecewise-constant bin membership along $D + \lambda S$, and full-rank coordinate invariance. Every empirical row now identifies which layer changed. See Section 2 and Appendix A.

M2. Degree of architecture heterogeneity

Response. Accepted. The revision separates point-estimate sign, a dependent-set sign test, and paired coefficient differences. Main Table 3 contains direct paired differences, intervals, and MDEs; all Q5–Q1 intervals contain zero. Native-support and BCC-grouping estimates are shown separately rather than pooled into one common-sample experiment. We no longer infer heterogeneity from individual significance patterns.

M3. Construct validity of external architectures

Response. The architecture matrix now records technology scope, primitive, label generation, publication and occupational-information vintages, horizon where verified, mapping, and the downstream hypothesis. OECD is described as a broader capability/demand and horizon construct, not a noisy LLM reliability measure. Webb AI and Webb software are distinguished. The external-admission rule is preserved as an operational screen, while the distinct-cut condition is explicitly representation-dependent rather than construct validity. The paper also reports unconditioned and Webb-conditioned BCC beta results, which are nearly identical for that binary grouping without making a general computerization claim. See Section 2/6 and Appendix A/E.

M4. Reference group and broad occupational variation

Response. This is now the substantive center of the paper. We replace "reference dependence" with exact descriptions of changed contrasts, retain the Q1/Q5 stock paths, add profile/equality/monotonicity tests, and estimate both SOC2-by-post and SOC2-by-month conditioned models with information and support diagnostics. The regression results, not the assumption-dependent permutation percentile, determine the framing. See Section 5 and Appendix C/E.

M5. Corrected pipeline and bridge uncertainty

Response. The repaired 113-month coefficient is now the substantive baseline; the frozen result is a chronology benchmark. We add a fail-closed taxonomy-compatibility gate, one-to-one-target estimates, split-source stock and information shares, age-allocation accounting ranges, and predeclared tilt scenarios. We do not claim the common-proportion bridge resolves age-specific allocation.

The 490/495 discrepancy is fully reconciled. The raw balanced universe is 539. The 490 support requires beta; the 495 support requires repaired AIOE plus Webb. They overlap in 466, with 24 beta-only and 29 AIOE-plus-Webb-only occupations. See Section 3, the sample-flow table, and Appendix B.

M6. Source of randomness and bootstrap mechanics

Response. Section 4 now states the target as a stochastic occupation-shock parameter conditional on realized CPS cells and constructed treatment. It lists household survey design, label generation, and mapping allocation as uncovered uncertainty. The nuisance-adjusted score, fitted information, cluster aggregation, finite-cluster factor, Rademacher multipliers, studentizer, critical value, and common-draw paired construction are described explicitly. The failed pseudo-outcome refit is correctly treated as evidence that that construction was inappropriate, neither validation nor invalidation of the score bootstrap. Draws increase from 999 to 9,999 for central revised inference; IUT and simultaneous-band claims remain distinct.

M7. Primitive units and F/G

Response. Accepted. Direct D, S and AIOE-plus-beta models are presented before the appendix rotation. Effective raw-unit centroid weights are reported, and leave-one-component changes are described as changes in implicit weights. F/G and A/E are explicitly the same fitted column space. See Section 6 and Appendix I.

M8. Mobility scale and benchmark

Response. Mobility is demoted and its claims narrowed. Thresholds are decision rules in standardized or rank units, not welfare magnitudes. The family-balanced matrix prevents multiplicity of near-duplicate measures from defining the result. The exact identity $\Delta X(\lambda) = \Delta D + \lambda \Delta S$ shows that beta cannot introduce a new zero-threshold sign conflict when alpha and broad agree. Benchmark and realized conflict are recomputed on identical represented support; omitted support is bounded. Household-linked sampling uncertainty is separated from Monte Carlo error, and at least two valid no-self repair distributions are compared. See Appendix H.

M9. Verifiable chronology without revision-history prose

Response. Revision-history phrases are removed from the main scientific narrative. The appendix preserves the dated protocol, protected-outcome definition, frozen and confirmatory tags, information set, hypothesis-to-test mapping, scripts, samples, seeds, and output hashes. We do not imply that repository hashes constitute external prospective registration. A result manifest links each headline number to its script and machine-readable output. See Appendix L and the reproducibility package.

Claims retired

The revised paper does not claim a causal AI employment effect, six independent validations, architecture equivalence, a monotone exposure response, a food-service mechanism, a CPS replication of proprietary ADP hiring results, survey-design-complete inference, or a coding-error correction inferred from occupation reversals. It also does not treat an invalid taxonomy merge as a defensible robustness specification.

Next revision pass

The two reports' secondary comments are preserved in the machine-readable response matrix. The next pass will focus on copyediting, citation alignment, remaining table conventions, and author-supplied affiliation/disclosure fields. None of those items will be marked complete merely because the major empirical revision is complete.